

GNN-Augmented BP Architectures for QLDPC Decoding

Jothiradithya Sai Konduru¹, Anish Chedalla¹, and Nithin Raveendran²

¹Paradise Valley High School, Phoenix, AZ, USA

²Department of Electrical and Computer Engineering, University of Arizona, Tucson, AZ, USA

Bivariate bicycle (BB) quantum low-density parity-check (QLDPC) codes combine finite encoding rate with low-weight checks, but their loopy Tanner graphs remain challenging for iterative decoding. ~~Learned message-passing can improve belief propagation (BP), although the measured gain depends strongly on the BP configuration used for training and evaluation. We study this issue using~~ We study a Tanner-graph neural network (TannerGNN) that produces qubit-wise log-likelihood corrections for ~~BP under a biased CSS component-noise model~~ belief propagation (BP) under Z -biased noise ($\eta = 20$), and ask how its gain depends on the decoder it is compared against. On the $[[72, 12, 6]]$ BB code, a model trained ~~with through~~ 10-iteration flooding BP reduces the logical error rate (LER) ~~from 0.0990 to 0.0838 on 8 of that decoder by $12.5\% \pm 0.6\%$ at $p = 0.04$ and by $10.6\% \pm 0.7\%$ at $p = 0.06$, outside the training range (five training seeds, 20,000 and 10,000 held-out syndromes at $p = 0.04$ and $\eta = 20$, a 15.4% relative reduction (paired McNemar $p < 10^{-15}$). This gain does not transfer to a stronger serial min-sum shots, every seed McNemar $p < 10^{-36}$). The gain comes mainly from helping BP converge, and it does not reach longer decoding: 100-iteration BP and serial BP-OSD decoder: the learned correction is statistically indistinguishable at lower tested error rates and degrades performance at $p \geq 0.04$. Independent baseline experiments further show that schedule, BP update rule, iteration count, and OSD configuration can both outperform GNN-augmented 10-iteration BP, and the same corrections make~~

~~BP-OSD slightly worse. A paired failure-set analysis shows why: the failure sets of BP and BP-OSD are nested, and most shots the GNN repairs are shots BP-OSD already decodes. Baseline experiments show that BP schedule and OSD post-processing change LER by orders of magnitude. These settings must therefore be specified and controlled when about $47\times$ on $[[288, 12, 18]]$, so these settings must be specified whenever learned decoders are compared. We also establish that prove that the hard decisions of unclipped min-sum BP is invariant, in its hard decisions, are invariant to a uniform positive rescaling of all channel LLRs within a CSS component. Thus, changing only the magnitude of a uniform prior cannot make the underlying rescaling of the channel LLRs, so a global noise estimate alone cannot make min-sum BP adaptive, although a learned network may still use global context through other pathways. Finally, synthetic experiments with heterogeneous. In contrast, revealing true per-qubit error rates show a large gap between mean-prior and oracle-prior decoding, motivating reduces BP-OSD's LER by 65–88% under heterogeneous code-capacity noise ($\sigma = 1$) and by 56% under circuit-level noise, identifying spatially resolved calibration side information rather than a single global noise scalar. The experiments show that an improvement over one BP baseline does not by itself establish an improvement over a stronger decoder, and that spatial calibration information is qualitatively different from a single global noise estimate as the more informative side information.~~

Nithin Raveendran: nithin@arizona.edu

1 Introduction

Bivariate bicycle (BB) codes provide a concrete finite-length setting in which QLDPC codes combine non-vanishing encoding rate, weight-six stabilizer checks, and syndrome-extraction circuits compatible with low-degree connectivity [1]. IBM’s current fault-tolerant architecture is explicitly based on BB codes, making the finite-length decoding behavior of this family practically relevant as well as theoretically interesting [2]. The canonical examples include the [72, 12, 6], [144, 12, 12], and [288, 12, 18] codes studied here.

The decoding bottleneck is not the absence of efficient iterative algorithms, but their finite-length failure modes. BP is attractive because its message updates are local and parallelizable, yet short cycles and degeneracy in QLDPC Tanner graphs create correlated messages and non-convergent or incorrect fixed points [3]. OSD post-processing can recover many BP failures, while recent learned decoders use graph neural networks (GNNs) either to replace BP or to intervene between BP stages [4, 5]. These approaches make the choice of baseline unusually consequential: a neural correction trained against one BP schedule or stopping rule need not address the residual failures of a stronger decoder.

A second issue is adaptivity. Hardware noise is neither spatially nor temporally uniform, and calibration information can therefore be useful to a decoder. However, three distinct sources of information should not be conflated: a global estimate of the physical error rate, spatially resolved per-qubit calibration data, and syndrome history. A decoder may respond differently to each. In particular, changing the magnitude of a uniform channel prior has a restricted effect on min-sum BP, whereas per-qubit priors can change the relative reliability ordering that drives both BP and OSD.

We study a GNN-augmented BP pipeline with two questions in view: *what gain does the learned correction provide relative to the BP decoder used in training, and what additional information is needed for noise adaptation?* We compare decoder configurations, training and evaluation baselines, and different forms of calibration information to isolate the source of the observed performance changes.

1.1 Contributions

Our main findings are:

1. **A held-out reproducible GNN gain over the training decoder, and its training-aligned BP baseline limits.** On [72, 12, 6] at $p = 0.04$ and $\eta = 20$, TANNERGNN + 10-iteration flooding BP reduces LER from 0.0990 to 0.0838 on 8,000 held-out syndromes, corresponding to a 15.4% relative reduction (McNemar $p < 10^{-15}$). The model is trained only on $p \in [0.02, 0.03]$, so this test is also outside the training error-rate range by $12.5\% \pm 0.6\%$ at $p = 0.04$ and $10.6\% \pm 0.7\%$ at $p = 0.06$ over five training seeds, with train, validation and test data drawn from independent seeds and checkpoints selected on validation data (Sec. 5.3). The corrected decoder remains weaker than 100-iteration BP and serial BP-OSD.
2. **Baseline configuration and alignment can dominate the apparent learned gain.** Serial min-sum BP with OSD post-processing is substantially stronger than the flooding BP used in GNN training. The training decoder and the evaluation decoder are part of the experimental condition. The same corrections do not improve serial BP-OSD and are significantly worse in 14 of 15 seed-error-rate pairs from $p = 0.03$ on. The same learned correction does not improve this stronger decoder and is worse at the two highest tested error rates. Separate experiments on failure sets of BP and BP-OSD are nested (93% of BP-OSD failures are BP failures), and 64–69% of the shots the GNN repairs are already decoded correctly by BP-OSD. On [288, 12, 18] show order-of-magnitude sensitivity to BP schedule, update rule, iteration count, and OSD settings. The training decoder and evaluation decoder are therefore part of the experimental condition rather than interchangeable implementations of “BP”, changing only the BP schedule and then adding OSD-CS-10 lowers LER by $8.8\times$ and $5.4\times$ (Secs. 5.1 and 5.4).
3. **Global prior scaling and spatial calibration carry different information.** We prove that hard-Hard decisions of unclipped min-sum BP are invariant to a uniform positive rescaling of the channel LLRs within

Table 1: Bivariate bicycle codes studied in this work, with generator polynomials from [1]. $\mathbf{H}_x = [\mathbf{A} \mid \mathbf{B}]$ and $\mathbf{H}_z = [\mathbf{B}^\top \mid \mathbf{A}^\top]$ over $\mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$.

Code	Name	ℓ	m	$A(x, y)$	$B(x, y)$
$[[72, 12, 6]]$	—	6	6	$x^3 + y + y^2$	$y^3 + x + x^2$
$[[144, 12, 12]]$	gross	12	6	$x^3 + y + y^2$	$y^3 + x + x^2$
$[[288, 12, 18]]$	two-gross	12	12	$x^3 + y^2 + y^7$	$y^3 + x + x^2$

a CSS component. This result limits adaptivity obtained only by changing a uniform BP prior; it does not rule out a learned network using a global noise scalar as context. In synthetic heterogeneous-noise experiments, providing the Providing true per-qubit rates to BP-OSD yields large oracle improvements, motivating spatially resolved calibration inputs. Because code size and operating point co-vary in these experiments, we do not attribute the cross-code trend to distance alone, by comparison, reduces LER by up to 88% at code capacity and by 56% at circuit level (Secs. 5.2 and 5.5, 5.5 and 5.7).

2 Background

2.1 CSS codes and bivariate bicycle construction

A CSS code [6, 7] is defined by matrices $\mathbf{H}_x, \mathbf{H}_z \in \mathbb{F}_2^{m \times n}$ with $\mathbf{H}_x \mathbf{H}_z^\top = \mathbf{0}$. The syndrome of an error (e_x, e_z) decomposes as $s_x = \mathbf{H}_x e_z$, $s_z = \mathbf{H}_z e_x$ over $\mathbb{F}_2$, so X - and Z -type errors are decoded independently. This CSS independence is structurally important for the results of Sec. 5.2.

Bivariate bicycle codes [1] are constructed over the group algebra $\mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$. Polynomials $A(x, y)$ and $B(x, y)$ yield circulant block matrices:

$$\mathbf{H}_x = [\mathbf{A} \mid \mathbf{B}], \quad \mathbf{H}_z = [\mathbf{B}^\top \mid \mathbf{A}^\top]. \quad (1)$$

We study three codes (Table 1).

2.2 Belief propagation and its failure modes

Min-sum BP on a CSS Tanner graph iterates variable-to-check and check-to-variable messages for T steps. For a check c_j with measured syn-

drome bit s_j and a neighboring variable v_i ,

$$\mu_{v_i \rightarrow c_j}^{(t)} = \Lambda_i + \sum_{j' \in \mathcal{N}(i) \setminus j} \mu_{c_{j'} \rightarrow v_i}^{(t-1)}, \quad (2)$$

$$\mu_{c_j \rightarrow v_i}^{(t)} = (-1)^{s_j} \prod_{i' \in \mathcal{N}(j) \setminus i} \text{sgn}(\mu_{v_{i'} \rightarrow c_j}^{(t)}) \times \min_{i' \in \mathcal{N}(j) \setminus i} |\mu_{v_{i'} \rightarrow c_j}^{(t)}|. \quad (3)$$

The channel LLR for qubit i under Z -biased noise is $\Lambda_{z,i} = \ln((1 - p_z)/p_z)$, $p_z = p\eta/(\eta + 1)$. CSS decoding separates X and Z components, processing each with its respective parity-check matrix.

The Tanner graphs of the finite-length BB codes considered here contain many short cycles. Message reuse around these cycles creates statistical dependencies that are absent on a tree and can lead BP to oscillate, converge to a syndrome-inconsistent estimate, or converge to the wrong logical class. We therefore treat short-cycle structure as a finite-length decoder mechanism to be measured, rather than attributing every cycle directly to the CSS orthogonality condition. Post-processing methods including ordered-statistics decoding (OSD) [8] recover many such failures but add $O(n^3)$ worst-case complexity. As shown in Sec. 5.1, the specific OSD configuration—schedule, base method, and iteration count—can change LER by more than two orders of magnitude on the same code. We therefore report these settings explicitly in the comparisons below.

2.3 Noise models

We use

Code capacity. Each data qubit independently suffers a Z flip with probability p_z and an X flip with probability p_x .

$$p_z = \frac{p\eta}{\eta + 1}, \quad p_x = \frac{p}{\eta + 1}, \quad \eta = 20. \quad (4)$$

No separate Y channel is sampled: a qubit carries a Y error only if it receives both flips, which happens with probability $p_x p_z \approx p^2/22$. This is therefore a Z -biased CSS component noise model with $\eta = 20$. [Confirm the simulator's treatment of Y errors. The equation

$$p_z = \frac{p\eta}{\eta + 1}, \quad p_x = \frac{p}{\eta + 1}.$$

For temporal drift experiments we apply three modulation models to the base rate independent X/Z model, not depolarizing noise. Syndromes are $\mathbf{s}_x = \mathbf{H}_x \mathbf{e}_z$ and $\mathbf{s}_z = \mathbf{H}_z \mathbf{e}_x$, and the two CSS components are decoded separately with priors p_z and p_x .

Spatial heterogeneity. For the oracle experiments each data qubit has its own rate $p_i = p e^{\sigma z_i - \sigma^2/2}$, $z_i \sim \mathcal{N}(0, 1)$, redrawn every shot and capped at 0.5, so the mean rate is $p \pm \text{sinusoidal } (p(t) = p + A \sin(2\pi t/T))$, for every contrast σ ; Eq. (4) is then applied per qubit.

Temporal drift. The training data for the GNN use a sinusoidal modulation of the base rate, $p(t) = p_0 + A \sin(2\pi t/T)$ with $A = 0.015$ and $T = 500$ shots. The pipeline also implements an Ornstein–Uhlenbeck model, $p_{t+1} = p_t + \theta(p_0 - p_t) + s \xi_t$ with $\theta = 0.1$, $s = 0.005$, $\xi_t \sim \mathcal{N}(0, 1)$, and random telegraph noise (RTN), all motivated by TLS physics switching between $p_0 \pm 0.01$ with probability 0.005 per shot; all rates are clipped to $[10^{-6}, 0.5]$. These models are motivated by fluctuating two-level-system defects [9, 10]. The decoder receives no direct access to instantaneous noise parameters in any experiment.

Circuit level. See Sec. 5.7 for the syndrome extraction circuit, detectors and circuit noise.

3 The ADAPTIVEBP Framework

3.1 TANNERGNN

TANNERGNN operates on the unified CSS Tanner graph, with edges distinguished by type $\tau \in \{X\text{-check}, Z\text{-check}\}$. Each variable node v_i is initialised with node receives a 4-dimensional feature vector $\mathbf{x}_i = [A_i, f_i, 1, 0]$, where f_i is the fraction of neighbouring checks that are violated; each check whose first entry is a value and whose last three entries one-hot encode the node type: a variable node v_i receives $\mathbf{x}_i = [\ell, 1, 0, 0]$, where $\ell = \ln((1-p)/p)$ is the prior LLR, and an X -check (Z -check) node c_j receives $\mathbf{x}_j = [0, s_j, 0, 1] [s_j, 0, 1, 0] ([s_j, 0, 0, 1])$, where s_j is its syndrome bit.

An input projection maps features to hidden dimension $d_h = 64$ via

$\mathbf{h}_i^{(0)} = \text{LayerNorm}(\mathbf{W}_{\text{proj}} \mathbf{x}_i) d_h$ ($d_h = 32$ for the models of Sec. 5.3) via $\mathbf{h}_i^{(0)} = \text{ReLU}(\text{LayerNorm}(\mathbf{W}_{\text{proj}} \mathbf{x}_i))$. Each

of $L = 3$ message-passing layers applies separate edge-type MLPs, optional attention weighting, and a GRU node update:

$$\mathbf{m}_i^{(\ell)} = \sum_{j \in \mathcal{N}(i)} \text{MLP}_{e, \tau(i, j)}^{(\ell)}([\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}]), \quad (5)$$

$$\mathbf{h}_i^{(\ell+1)} = \text{GRU}(\mathbf{h}_i^{(\ell)}, \mathbf{m}_i^{(\ell)}). \quad (6)$$

A readout MLP maps each variable node’s final hidden state to a per-qubit LLR correction: $\Delta_i = \text{MLP}_{\text{read}}(\mathbf{h}_i^{(L)})$.

Remark 1 (Architectural limitation under biased noise). The current implementation applies the same scalar Δ_i to both $\Lambda_{z, i}$ and $\Lambda_{x, i}$, and computes GNN input features from their average single prior LLR. Under $\eta = 20$, $\Lambda_{z, i}$ and $\Lambda_{x, i}$ differ substantially (factor ≈ 1.7 in LLR magnitude at $p = 0.04$), so a single correction cannot simultaneously be optimal for both error types. Separate Z - and X -readout heads provide a direct way to remove this coupling and should be tested under the same biased-noise conditions.

3.2 FiLM conditioning

Feature-wise Linear Modulation (FiLM) [11] conditions each GNN layer on an external context. A small generator network maps the syndrome-weight estimate $\hat{p} = \|\mathbf{s}\|_1 / (m_x + m_z)$ to per-layer affine parameters $(\gamma^{(\ell)}, \beta^{(\ell)}) = \text{FiLMGen}(\hat{p})$, modulating hidden activations as $\tilde{\mathbf{h}}^{(\ell)} = \gamma^{(\ell)} \odot \mathbf{h}^{(\ell)} + \beta^{(\ell)}$. This conditioning pathway is designed to let a single model respond to global changes in the noise level without retraining. It should not be confused with spatial calibration: $\hat{p}$ is one scalar shared by the entire graph and therefore contains no direct per-qubit reliability information. Section 5.2 establishes a narrower result: uniform rescaling of the channel prior cannot change the hard decisions of the underlying unclipped min-sum BP decoder. That invariance does not, by itself, imply that a GNN cannot use a global scalar as contextual information.

3.3 Neural BP and interleaved pipeline

Differentiable BP generalises (2)–(3) with learnable per-iteration scaling weights

PLACEHOLDER: fig_tannergnn_arch (keep your Overleaf copy)

Figure 1: TANNERGNN architecture. Variable and check nodes exchange typed messages over $L = 3$ layers via edge-type-specific MLPs; a GRU update aggregates messages at each node. In the FiLM variant, FiLM modulation (not shown) scales the hidden activations with syndrome-derived conditioning at each layer; the model of Sec. 5.3 does not use FiLM.

Algorithm 1: Interleaved GNN-BP De-coding

Input: $\mathbf{s}$, channel LLR Λ , noise estimate $\hat{p}$
Output: Error estimate $\hat{\mathbf{e}}$
 $(\gamma, \beta) \leftarrow \text{FiLMGen}(\hat{p})$ // Conditioning
 $\Delta^{(1)} \leftarrow \text{TANNERGNN}(\Lambda, \mathbf{s}, \gamma, \beta)$
 $\Lambda' \leftarrow \Lambda + \Delta^{(1)}$ // Prior correction
 $\Lambda_{\text{mid}} \leftarrow \text{NeuralBP}(\Lambda', \mathbf{H}, T_1)$
 $p_i^{\text{mid}} \leftarrow \sigma(-\Lambda_{\text{mid},i}) \forall i$
 $\Delta^{(2)} \leftarrow \text{TANNERGNN}(\Lambda_{\text{mid}}, p^{\text{mid}}, \mathbf{s}, \gamma, \beta)$
 $\Lambda'' \leftarrow \Lambda_{\text{mid}} + \Delta^{(2)}$ // Mid correction
 $\Lambda_{\text{final}} \leftarrow \text{NeuralBP}(\Lambda'', \mathbf{H}, T_2)$
 $\hat{\mathbf{e}}_i \leftarrow \mathbb{1}[\Lambda_{\text{final},i} < 0] \forall i$
return $\hat{\mathbf{e}}$

$w_{\text{ch}}^{(t)}, w_{\text{vtr}}^{(t)}, w_{\text{ctv}}^{(t)} = \text{softplus}(w_{\text{raw}}^{(t)})$ (softplus ensures positivity; $w_{\text{raw}} = 0$ at initialisation gives $w \approx \ln 2 \approx 0.693$, recovering near-standard BP).

Algorithm 1 gives the interleaved pipeline. The GNN intervenes twice: before BP to correct channel priors, and after T_1 iterations using the intermediate posterior information. Qubits with $p_i^{\text{mid}} \approx 0.5$ have low posterior confidence and can receive a second learned correction. Both correction stages are trained jointly through the differentiable BP unroll.

3.4 Training

~~Training uses~~

Training loss. For a shot with prior LLR $\ell = \ln((1-p)/p)$ on every qubit, the GNN outputs a correction δ_j for each qubit j , giving the corrected error probability $q_j = \sigma(-(\ell + \delta_j))$. The model is trained on the focal loss [12] against

Table 2: Architecture—Configuration of the TANNERGNN model behind Sec. 5.3 and training hyperparameters Sec. 5.4.

<i>Architecture</i>	
Message-passing layers / hidden width	3 / 32
Node features	4 (value, one-hot node type)
Residual connections, layer norm	yes
FiLM / attention	not used
Parameters	39,329
<i>Training</i>	
Data	8,000 shots, sinusoidal drift, $p \in [0.005, 0.045]$
Loss	focal, $\alpha = 0.25$, $\gamma = 2$, on $\mathbf{e}_z$
Optimiser	AdamW, lr 2×10^{-3} , weight decay 10^{-4}
Batch size / epochs	64 / 20, gradient clipping 1.0
Checkpoint	lowest failures on 4,000 validation shots
Training seeds	5
<i>Decoders</i>	
BP (training and Table 4)	flooding min-sum, scaling 0.8, 10 iter.
BP-OSD reference	serial min-sum, scaling 0.8, 100 iter., OSD-CS-10

the true Z error $\mathbf{e}_z$,

$$\mathcal{L} = -\frac{1}{Bn} \sum_{b=1}^B \sum_{j=1}^n \alpha_t (1 - q_t)^\gamma \log q_t, \quad (7)$$

where $q_t = q_j$ and $\alpha_t = \alpha$ if $e_{z,j} = 1$, and $q_t = 1 - q_j$ and $\alpha_t = 1 - \alpha$ otherwise, with $\alpha = 0.25$, $\gamma = 2.0$ for class-imbalanced qubit errors, together with a syndrome consistency term. The noise range is experiment dependent; in particular, the positive held-out result of $\alpha = 0.25$ and $\gamma = 2$. No syndrome-consistency term is used for this model. At decoding time the corrected LLRs $\ell + \delta_j$ are passed to both CSS components. The earlier FiLM and interleaved models additionally used a syndrome-consistency term, the binary cross-entropy between the measured syndrome and $\frac{1}{2}(1 - \prod_{j \in \mathcal{N}(c)} (1 - 2q_j))$ for each check c (weight 0.1 in the configuration of the previous version).

The training data and splits of the headline experiment are described in

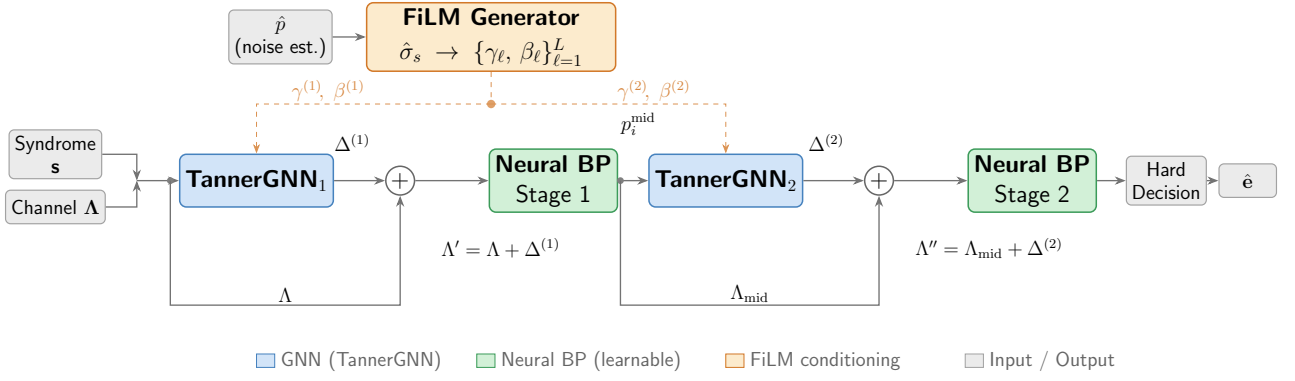


Figure 2: Interleaved GNN-BP pipeline. The GNN applies two successive LLR corrections: one before BP initialisation and one after T_1 iterations, using the mid-decoding posterior as an additional input. Both stages are trained end-to-end through the BP unroll.

Sec. 5.3 uses training rates $p \in [0.02, 0.03]$.
~~[Reconcile this with the earlier training configuration Lpr = Uniform(0.01, 0.10) and report the Data Range, number~~
~~Key hyperparameters are summarized; the~~
~~configuration is summarised in Table 2.~~

4 Experimental Setup

4.1 Evaluation regimes

Two distinct evaluation regimes are used throughout this paper, and the distinction is central to interpreting the results:

Regime A (training-aligned baseline): The GNN is evaluated against flooding BP, the base decoder used during training. This comparison isolates the gain relative to the decoder represented in the training computation graph.

Regime B (stronger classical baseline): The same learned correction is evaluated with the serial-schedule min-sum BP-OSD reference configuration. This tests whether the correction also improves the stronger classical decoder considered here; Sec. 5.4 shows that it does not in the tested regime.

Unless otherwise stated, all experiments use Z -biased noise with $\eta = 20$.

4.2 Statistical protocol

Logical failure. For each shot let $\hat{\mathbf{e}}_z$ and $\hat{\mathbf{e}}_x$ be the decoder’s hard decisions and let $\mathbf{r}_z = \mathbf{e}_z \oplus \hat{\mathbf{e}}_z$ and $\mathbf{r}_x = \mathbf{e}_x \oplus \hat{\mathbf{e}}_x$ be the residual errors. A shot is counted as a logical failure if either (i) the estimate does not reproduce the measured syndrome, $\mathbf{H}_x \hat{\mathbf{e}}_z \neq \mathbf{s}_x$ or $\mathbf{H}_z \hat{\mathbf{e}}_x \neq \mathbf{s}_z$; or (ii) the syndrome is reproduced but the residual acts

nontrivially on the code space, $\mathbf{L}_x \mathbf{r}_z \neq \mathbf{0}$ or $\mathbf{L}_z \mathbf{r}_x \neq \mathbf{0}$ for $\mathbf{L} \in \{0, \mathbf{I}, \mathbf{X}, \mathbf{Z}\}$. We report the Data Range, number span the $k = 12$ independent X -type (Z -type) logical operators modulo stabilizers. Condition (i) is required because a syndrome-inconsistent residual is not a logical operator, so its parity against a fixed set of logical representatives is not meaningful; non-converged BP outputs are therefore always counted as failures. The X - and Z -components are combined by logical OR, so the reported LER is the probability that at least one of the 12 encoded qubits suffers a logical error of any Pauli type in a single shot. At circuit level the same rule is applied to the detector error model: a shot fails if the decoded fault set does not reproduce the observed detection events, or if it does and its predicted observable flips differ from the true ones.

For absolute LER comparisons we treat operating points with fewer than 100 observed logical-error events as preliminary; this is a reporting heuristic rather than a formal power calculation. Decoder pairs are evaluated on *identical* syndrome samples, enabling paired McNemar tests [13]. We report Wilson score 95% confidence intervals for LER estimates.

~~Random seeds are fixed at $s = 42$ for syndrome generation and $s + 999$ for validation splits. Unless otherwise stated, the reported Training, validation and test data are drawn with distinct random seeds, so no evaluation shot shares a random stream with a training shot, and learned-decoder result is based on a single training seed; its confidence interval and McNemar test therefore quantify test-set sampling uncertainty conditional on that trained checkpoint, not~~

Figure 3: Experimental evaluation pipeline. Syndromes are sampled under static or drift noise models; each decoder receives identical syndrome batches to enable McNemar paired testing. Oracle experiments substitute true per-qubit rates for the uniform mean prior.

training-to-training variability results are reported over five training seeds (initialisation and minibatch order). All training and evaluation used CPU only; no GPU was available for this work. Representative per-epoch training times are ~ 17 s (154k-parameter GNN-BP) and ~ 37 s (39k-parameter model of Sec. 5.3 takes about 16 s (183k-parameter interleaved model) on four CPU cores.

5 Results

5.1 Decoder configuration sensitivity

Before presenting the GNN results, we quantify the sensitivity of the classical baseline to decoder configuration. This sensitivity sets the scale against which the learned-decoder results must be interpreted.

Table 3 illustrates how strongly decoder configuration changes the finite-length result. In the 5,000-shot pilot, the original parallel on 100,000 identical shots. Parallel product-sum BP-OSD configuration reduces LER from 0.0506 to 0.0346 relative to with 30 iterations lowers the LER of the corresponding plain BP, but remains far worse than the serial min-sum configurations. The reference only from 0.0569 to 0.0398, whereas serial min-sum BP-OSD produces only one logical error in this sample, so the value 0.0002 should be interpreted as a preliminary estimate rather than a precisely resolved LER. BP with 100 iterations reaches 0.0017 without OSD and 0.00034 with it, more than two orders of magnitude below the parallel product-sum configurations.

A separate $N = 100,000$ 100,000-shot experiment is shown in (Fig. 4. In that experiment,

changing to) changes one factor at a time. The flooding comparator is min-sum BP with scaling factor 0.8, at most 100 iterations, stopping at the first iteration whose hard decision satisfies the syndrome, with a parallel (flooding) schedule; the second decoder is identical except for the serial schedule and then adding; the third adds OSD-CS-10 yields an overall approximately $40\times$ reduction relative to the flooding comparator. [State the exact BP method, scaling factor, iteration count, . The logical error rate falls from 1.28×10^{-2} to 1.45×10^{-3} ($8.8\times$, paired $n_{10}/n_{01} = 1138/3$) and then to 2.7×10^{-4} ($5.4\times$, $118/0$), about $47\times$ overall.

These data do not identify a universally optimal configuration. They do show that schedule, update rule, iteration budget, and post-processing can dominate the measured finite-length LER. Learned-decoder comparisons must therefore specify and control these parameters.

5.2 Uniform-prior scaling invariance of min-sum BP

Proposition 1 (Uniform positive-scaling invariance). *Consider separate-CSS min-sum BP with no message clipping or offset. If all channel LLRs in one CSS component are multiplied by the same factor $a > 0$, then every BP message and posterior LLR is multiplied by a at every iteration. Consequently, all hard decisions are unchanged.*

Proof. At initialization, each variable-to-check message is proportional to the corresponding channel LLR and therefore scales by a . Assume all variable-to-check messages at iteration t have scaled by a . The syndrome factor $(-1)^{s_j}$ in Eq. (3) is unchanged, the sign of every message is unchanged because $a > 0$, and $\min_i a|x_i| =$

Table 3: Effect of BP-OSD configuration on $[[288, 12, 18]]$, $p = 0.04$, $\eta = 20$, ~~5,000–100,000~~ identical shots, ~~separate-CSS decoding with priors $p_Z = p\eta/(\eta + 1)$, $p_X = p/(\eta + 1)$~~ . ~~Cl~~~~s~~ ~~95% Wilson intervals~~ in brackets. ~~OSD is OSD-CS of order 10.~~

Config	Schedule	Method	Iter	LER	Errors
Original BP	parallel	prod-sum	30	0.0506–0.0569 [0.045, 0.0570.0555, 0.0583]	253–5,689
Reference BP	serial	ms(0.8)	100	0.0014–0.00174 [0.001, 0.0030.00150, 0.00202]	7–174
Original BP-OSD	parallel	prod-sum	30	0.0346–0.0398 [0.030, 0.0400.0386, 0.0411]	173–3,983
Ref. BP-OSD	serial	ms(0.8)	100	0.0002–0.00034 [0.000, 0.0010.00024, 0.00048]	1–34

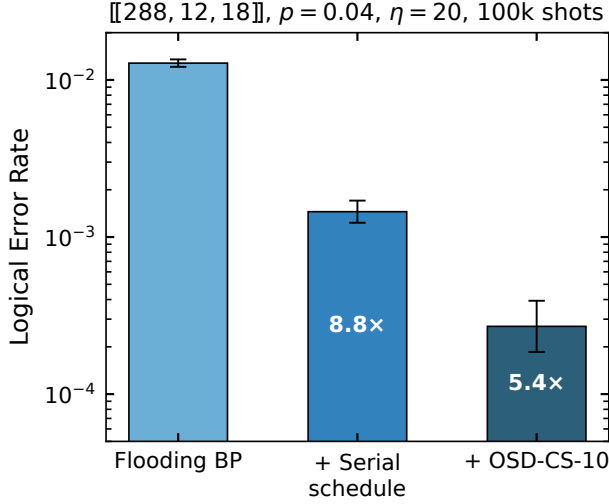


Figure 4: Baseline decomposition on $[[288, 12, 18]]$, $p = 0.04$, $\eta = 20$, 100k ~~identical~~ shots. ~~The figure is intended to isolate the effect of serial scheduling Min-sum BP (scaling 0.8, at most 100 iterations) with flooding and subsequent OSD-CS-10 then serial schedule, then serial BP-OSD-CS-10. Bars: LER with 95% Wilson intervals; labels give the exact flooding-BP comparator configuration must be reported in the text before the multiplicative factors are interpreted causally factor gained at each step.~~

$a \min_i |x_i|$. Hence every check-to-variable message also scales by a . Equation (2) then shows that the next variable-to-check messages scale by a . Induction establishes the claim for all iterations and for the final posterior LLRs. $\square$

For a uniform channel prior, replacing a common positive LLR Λ by another positive value Λ' is equivalent to the scaling $a = \Lambda'/\Lambda$. The proposition explains why changing only the assumed uniform physical error rate cannot change hard decisions of ideal min-sum BP. Clipping, offsets, damping rules that depend on magnitude, non-uniform priors, or sum-product updates can break the exact scaling argument.

We ~~observed the expected invariance in the~~

~~implemented decoder: 1,740/1,740 bit-identical outputs when checked the proposition on 2,000 shots each. With ldpc min-sum BP (serial or flooding, 100 iterations), changing the assumed prior-uniform rate from $p = 0.015$ to $p = 0.0286–0.0286$ on $[[288, 12, 18]]$, and 1,000/1,000 bit-identical outputs for $p = 0.04$ versus $p = 0.02$ in the tested or from 0.04 to 0.02 on $[[72, 12, 6]]$, leaves every hard decision unchanged (2,000/2,000), and an assumed-bias sweep $\eta \in \{1, 2, 5, 10, 20, 50, 100\}$ at $p = 0.03$ gives identical decisions at every value. Two cases fall outside the proposition, and the invariance fails there: our PyTorch min-sum setting. The same implementation clamps messages to ± 20 and changes 18/2,000-shot test was also identical for the tested 000 decisions on $[[288, 12, 18]]$, and sum-product configuration; because Proposition 1 does not apply to the tanh nonlinearity, we report that observation as empirical rather than structural. BP changes 25–43 of 2,000.~~

Remark 2 (What the proposition does and does not imply). *The result applies to the BP pathway when adaptivity enters only by changing a uniform channel-prior magnitude. It does not establish that a global conditioning scalar is information-theoretically useless to a GNN: a learned model could combine a global noise estimate with local syndrome structure and change its nonlinear decision rule. The experiments below test a narrower hypothesis: spatially resolved per-qubit calibration can change the relative reliabilities seen by the decoder, whereas a single graph-wide scalar cannot provide that spatial information.*

5.3 GNN improvement over flooding BP

We evaluated the GNN against flooding BP—the BP, the decoder it was trained ~~against on~~ ~~against~~. The training set is 8,000 ~~held-out~~ syndromes. ~~shots from two sinusoidally~~

~~GNN-BP vs. flooding BP on $\llbracket 72, 12, 6 \rrbracket$, $\eta = 20$, 8,000 held-out test syndromes. GNN-trained for 20 epochs on separate 8,000 training syndromes at $p \in [0.02, 0.03]$. Best checkpoint at epoch 5 by validation LER.~~

Table 4: ~~GNN-BP vs. flooding BP on $\llbracket 72, 12, 6 \rrbracket$, $\eta = 20$, five training seeds (mean $\pm$ s.d.). Failures follow the definition in Sec. 4.2. 95% Wilson intervals for single decoders are ± 0.004 at $p = 0.04$ and ± 0.009 at $p = 0.06$.~~

Decoder	$p = 0.04$ (20k shots)	$p = 0.06$ (10k shots)
BP, 10 iter.	0.1077	0.2867
GNN + BP, 10 iter.	0.0942 ± 0.0006	0.2564 ± 0.0019
relative reduction	$12.5\% \pm 0.6\%$	$10.6\% \pm 0.7\%$
McNemar n_{10}/n_{01} (range)	313–370 / 64–95	343–394 / 66–80
BP, 100 iter.	0.0863	0.2403
Serial BP-OSD-CS(10), 100 iter.	0.0707	0.2138

~~drifting streams, $p(t) = p_0 + 0.015 \sin(2\pi t/500)$ with $p_0 \in \{0.02, 0.03\}$, so the training rates span $p \in [0.005, 0.045]$. Validation (4,000 shots at $p = 0.04$) and test sets (20,000 shots at $p = 0.04$, inside the training range; 10,000 shots at $p = 0.06$, outside it) are drawn with separate random seeds. The checkpoint is chosen by the lowest validation failure count over 20 epochs, each test set is scored once on that checkpoint, and training is repeated with five seeds (Table 2).~~

~~Table 4 gives the primary learned-decoder result. At the epoch-5 checkpoint, GNN-BP reduces LER by 15.4% relative to flooding BP on 8,000 held-out syndromes. GNN pre-processing reduces the failure rate of 10-iteration BP by $12.5\% \pm 0.6\%$ at $p = 0.04$ ($n_{10} = 170, n_{01} = 48$, McNemar $\chi^2 = 67.16, p < 10^{-15}$). The test point and $10.6\% \pm 0.7\%$ at $p = 0.06$, which lies outside the training range $p \in [0.02, 0.03]$, so the result demonstrates transfer to a higher physical error rate for this checkpoint. Because only one training seed is reported, we do not interpret the paired test as evidence for training-to-training reproducibility. A: every seed is significant ($p < 10^{-36}$, McNemar). The gain comes mostly from convergence: shots where BP fails to reproduce the syndrome fall from 1,740 to 838–975 of 20,000. The gain does not close the gap to longer decoding: 100-iteration BP (0.0863) and serial BP-OSD (0.0707) both outperform GNN + 10-iteration BP.~~

~~On the original single-seed checkpoint, a gradient audit found non-zero signal through the network (input-to-readout gradient ratio 0.22) and non-trivial LLR corrections (mean $|\Delta_i| = 0.854$).~~

Table 5: ~~GNN-BP~~ GNN+OSD vs. serial BP-OSD on $\llbracket 72, 12, 6 \rrbracket$, $\eta = 20$, ~~8,000~~ 20,000 identical shots ~~per p , five GNN training seeds. Decoders evaluated on identical syndromes~~ Serial min-sum BP-OSD-CS-10 (scaling 0.8, 100 iterations).

p	Serial BP-OSD	GNN+OSD (5 seeds)	Seeds worse ($p_{\text{McN}} < 0.05$)
0.02	0.0093	0.0095–0.0102	1/5
0.03	0.0297	0.0306–0.0317	5/5
0.04	0.0676	0.0688–0.0709	4/5
0.05	0.1354	0.1384–0.1418	5/5

~~A fixed-dataset capacity test provides an additional diagnostic: a GNN trained on 1,000 fixed syndromes for 50 epochs achieved a 14% LER reduction reduced the failure count on those same syndromes (McNemar $p = 0.004$, $n_{10} = 17$ from 105 to 93 (11%; $n_{10} = 15$, $n_{01} = 3$, McNemar $p = 0.0095$). This experiment tests the ability of the architecture and optimizer to fit the supplied examples; it is not a generalization result. The interleaved model reached its best validation checkpoint at epoch 1 and did not train successfully, so no performance conclusion about that architecture is drawn from the current run (see Limitations) architecture of Sec. 3.3 is not evaluated in this work.~~

~~Interleaved model training curves (183k parameters). The best validation checkpoint occurs at epoch 1 and the validation loss increases thereafter. The current run does not provide a trained interleaved decoder for comparison.~~

5.4 Training/evaluation baseline mismatch

~~Although the GNN improves flooding BP, the same correction does not improve the serial min-sum BP-OSD reference configuration. Note that Tables 4 and 5 report different evaluation regimes (Regime A and Regime B of Sec. 4.1) on different syndrome sets, and their LER values are not directly comparable. We pass the corrected LLRs of each of the five trained models to BP-OSD and compare with BP-OSD alone on 20,000 identical fresh shots per error rate (Table 5); these shots differ from the test sets of Table 4.~~

~~Table 5 and Fig. 5 show that the GNN-corrected decoder is statistically indistinguishable from never improves on serial BP-OSD at the two lower error rates and: at $p = 0.02$ only one seed of five is significantly worse, and from $p = 0.03$ on 14 of 15 seed-error-rate pairs are significantly worse at~~

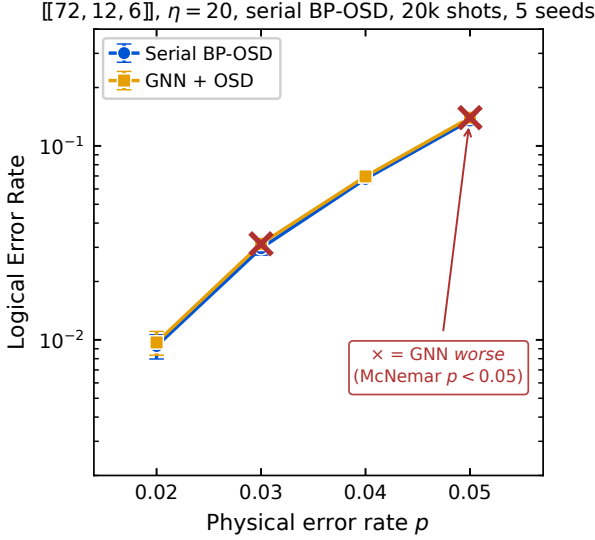


Figure 5: GNN+OSD vs. serial BP-OSD LER on $[[72, 12, 6]]$, $\eta = 20$, 20,000 shots per point; GNN+OSD is the mean over five training seeds. Red $\times$ marks operating points where GNN every seed is statistically worse (McNemar $p < 0.05$). Wilson 95% CI error bars.

the two higher rates. The same checkpoint improves checkpoints improve flooding BP, so these data show that the learned correction does not transfer unchanged to the stronger decoder.

~~The experiment establishes a transfer failure—~~

Table 6 compares the failure sets of all decoders on identical shots. The failure sets are nested rather than disjoint: every shot that 100-iteration BP fails is also failed by 10-iteration BP, and 93% of BP-OSD’s failures are BP failures. What separates the decoders is mainly convergence: 81% of 10-iteration BP failures are shots on which BP does not reach a syndrome-consistent estimate, whereas BP-OSD always returns one. No decoder fails on any error of weight $\leq 2 = (d-1)/2$; all failures involve errors of weight at least 3 (mean weight 5.4–5.7). In 1, but it does not by itself identify the residual-failure mechanism. The 411 of BP-OSD’s 1,414 failures the returned correction is syndrome-consistent and no heavier than the true error, so these failures are limited by the code and the noise, not by a decoder returning an unnecessarily heavy estimate.

The GNN’s gains lie where BP-OSD already succeeds. Of the 313–370 shots the GNN fixes for 10-iteration BP, 91–93% are shots on which BP did not converge, and only 31–36% are shots that BP-OSD fails; conversely, GNN+BP still fails

Table 6: Paired failure sets on $[[72, 12, 6]]$, $p = 0.04$, $\eta = 20$, 20,000 identical held-out shots. BP: 10-iteration flooding min-sum (the GNN’s training decoder). GNN rows give the range over five training seeds. “Non-conv.”: the estimate does not reproduce the syndrome.

Decoder	Failures	Non-conv.
BP, 10 iter.	2,155	1,740
BP, 100 iter.	1,727	700
Serial BP-OSD	1,414	0
GNN + BP, 10 iter.	1,874–1,906	838–975
GNN + BP-OSD	1,439–1,451	0
<i>Overlaps</i>		
BP-OSD failures that BP also fails	1,319 / 1,414 (93%)	
BP-100 failures that BP also fails	1,727 / 1,727 (100%)	
Shots GNN fixes for BP (BP fails, GNN+BP succeeds)	313–370	
of which BP did not converge	91–93%	
of which BP-OSD also fails	31–36%	
BP-OSD failures that GNN+BP also fails	87–88%	
GNN+BP-OSD vs. BP-OSD: newly failed / rescued	237–282 / 206–254	
newly failed shots that BP also fails	91–95%	

87–88% of BP-OSD’s failures. When the same corrections are passed to BP-OSD, they reshuffle rather than remove its failures (237–282 newly failed vs. 206–254 rescued shots), almost entirely among the hard shots on which BP itself fails. They also make OSD return heavier incorrect estimates (20–43 failures with an estimate heavier than the true error, vs. 3 for BP-OSD alone). The same pattern holds at $p = 0.06$ (10,000 shots): 96% of BP-OSD failures are BP failures, and 30–36% of the GNN’s fixes are BP-OSD failures.

The GNN was optimized through a 10-iteration flooding-BP computation graph and is then inserted ahead of a decoder with a different schedule, iteration budget, and OSD post-processing. The experiment therefore does not establish that LLR corrections learned through the flooding-BP computation graph remain useful after the schedule, iteration budget, and post-processing are changed. At the same time, claims that the two decoders fail on “structurally different” trapping sets require an explicit paired failure-overlap or trapping-set analysis, which is not included here. If available, add the Jaccard/overlap statistics of the two fa

Training directly against the target decoder, or against a differentiable surrogate of its post-processing stage, remains the appropriate test of whether learned augmentation can improve BP-OSD.

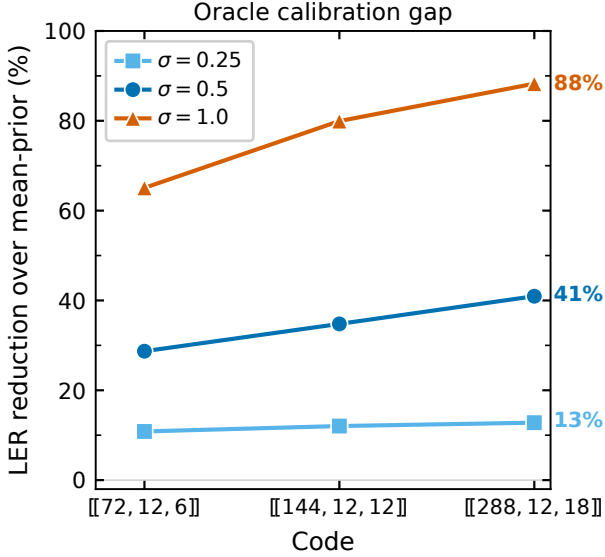


Figure 6: Oracle calibration gap (percentage LER reduction ~~from per-qubit vs. of per-qubit-oracle~~ over mean-prior BP-OSD) at $\sigma = 1.0$, 50k identical shots ($[[72, 12, 6]]$, $[[144, 12, 12]]$) and 30k shots ($[[288, 12, 18]]$) per point. Operating points: $p = 0.04$, 0.06, 0.07 respectively for the three codes; cross-code comparisons are partially confounded by the differing operating points.

5.5 Per-qubit calibration gap

Under spatially non-uniform per-qubit noise ($p_i \sim \text{LogNormal}(\log p_{\text{mean}}, \sigma)$, ~~fresh per shot~~), ~~a each data qubit has rate $p_i = p e^{\sigma z_i - \sigma^2/2}$ with $z_i \sim \mathcal{N}(0, 1)$ drawn fresh every shot (capped at 0.5), so the mean rate is p for every σ ; the mean-prior decoder uses the uniform rate p .~~ A decoder knowing the true per-qubit rates ~~substantially outperforms a (serial min-sum BP-OSD-CS-10 with the true p_i as priors) substantially outperforms the~~ mean-prior decoder.

Table 7 and Fig. 6 show a clear increase with noise contrast σ for each tested code. The reported percentage reduction is also larger for the larger-code rows, reaching 87%–88% on $[[288, 12, 18]]$ at $\sigma = 1.0$; however, the physical error rate is changed across codes ($p = 0.04$, 0.06, and 0.07). ~~Consequently, these data~~ At a common $p = 0.04$, $[[144, 12, 12]]$ shows larger gaps than $[[72, 12, 6]]$ at every σ (Table 7, last row), but $[[288, 12, 18]]$ fails too rarely at that rate to resolve the trend. These data therefore do not isolate code distance as the cause of the cross-code trend.

This experiment isolates the value of synthetic

per-qubit rate side information; it does not require machine learning. We use the true-rate decoder as an *oracle reference*, not as a formal upper bound on all possible decoders. The comparison quantifies how much the tested BP-OSD implementation can benefit when the realized per-qubit rates are revealed. The gap is *not* recoverable from a single syndrome shot under i.i.d. per-qubit noise: with a fresh field redrawn every shot, the uniform mean prior is Bayes-optimal within the class of estimators that observe only the syndrome, so no syndrome-conditioned GNN can do better than the mean prior in this regime. ~~A GNN that attempts to estimate per-qubit rates from a single syndrome is significantly worse than the mean prior at $\sigma = 1.0$ (GNN LER = 0.069 vs. mean LER = 0.064, McNemar $p = 2.7 \times 10^{-4}$) not because the architecture is broken, but because the task is information-theoretically infeasible.~~

5.6 Drift adaptation

Under persistent OU drift ($\sigma = 1.0$, $W = 32$ -round window, $[[72, 12, 6]]$, $p = 0.04$, 6,000 shots), the GNN beats stale calibration only at the highest tested drift volatility.

Table 8 and Fig. 7 show that the GNN beats stale calibration at vol= 0.5 ($p = 0.005$, 158 vs. 111 discordant pairs), but is beaten by the non-learned empirical frequency estimator at every volatility level (GNN vs. empir: $p = 2.7 \times 10^{-8}$ at vol= 0.5, GNN worse). Under independent per-qubit drift, direct per-qubit syndrome frequency counting captures most of the available signal, leaving little room for a learned estimator. Whether spatial correlation structure between qubits would shift this result is an open question.

5.7 Circuit-level evaluation

~~{Specify the complete circuit-level noise model: syndrome ex~~

Circuit and noise model. We simulate a Z -basis memory experiment on $[[72, 12, 6]]$ with 6 rounds of syndrome extraction. Data qubits are prepared in $|0\rangle$; each round measures all 36 X -checks and then all 36 Z -checks with dedicated ancillas, using CNOT layers from a greedy conflict-free schedule; the data qubits

Table 7: Oracle gap: percentage LER reduction ~~from per-qubit of per-qubit-oracle~~ over mean-prior ~~BP-OSD~~^{serial} BP-OSD-CS-10, 50,000 identical shots per cell. Parentheses: mean-prior failures. All McNemar $p \ll 10^{-3}$ $p < 10^{-20}$ in the upper three rows and $p < 10^{-5}$ in the last. ~~Parenthetical counts are mean-prior error events~~ The upper three rows use the operating points of Fig. 6; the last row fixes $p = 0.04$.

Code (d)	p	$\sigma = 0.25$	$\sigma = 0.5$	$\sigma = 1.0$
$[[72, 12, 6]]$ (6)	+11% (3457) 0.04	10.8% (3508)	+30% (3420) 28.7% (3432)	+66% (2988) 65.1% (3438)
$[[144, 12, 12]]$ (12)	+16% (2760) 0.06	12.0% (3165)	+38% (2683) 4.8% (3203)	+74% (2604) 79.9% (2936)
$[[288, 12, 18]]$ (18)	+18% (9220) 0.07	12.8% (3338)	+48% (5244) 40.9% (3307)	+87% (1750) 88.3% (2786)
$[[144, 12, 12]]$ (12)	0.04	19.8% (324)	45.5% (275)	85.3% (278)

Table 8: LER under per-qubit OU drift by volatility (vol). “Empir”: non-learned per-qubit frequency estimator.

vol	Oracle	Stale	Empir	GNN	Verdict
0.00	0.033	0.033	0.057	0.064	GNN worse, $p = 5 \times 10^{-36}$
0.10	0.031	0.038	0.057	0.067	GNN worse, $p = 8 \times 10^{-30}$
0.35	0.005	0.027	0.038	0.050	GNN worse, $p = 5 \times 10^{-19}$
0.50	0.004	0.052	0.035	0.044	GNN beats stale, $p = 0.005$

are finally measured in the Z basis. Detectors are (i) the 36 Z -checks of round 1, which are deterministic for the prepared state, (ii) the XOR of consecutive measurements of every check in rounds 2–6, and (iii) the XOR of the last Z -check outcome with the parity of the measured data qubits in its support, giving 432 detectors and 12 logical observables (the Z -type logical operators). Noise is a biased single-qubit Pauli channel with $p_X = p/(\eta + 1)$, $p_Z = p\eta/(\eta + 1)$, $p_Y = 0$, $\eta = 20$, applied to every qubit after each one- and two-qubit gate and to every idle qubit at each time step; measurements and resets flip with probability p . Under spatially non-uniform noise each qubit q (data and ancilla) has rate $p_q = p_{\text{mean}} e^{\sigma^2 z_q}$, $z_q \sim \mathcal{N}(0, 1)$, $\sigma = 1$, on all of its fault locations.

Decoders. We decode the detector error model (DEM; 4,068 fault mechanisms) with BP-OSD (min-sum, scaling 0.625, serial schedule, 100 iterations, OSD-CS order 10). The mean-prior decoder uses a uniform rate equal to the log-normal mean $p_{\text{mean}} e^{\sigma^2/2}$; the oracle uses the true p_q . We draw 20 (40) rate profiles at $p_{\text{mean}} \leq 0.003$ (0.006) and 500 shots per profile. A shot fails if the decoded fault set does not reproduce the detection events or predicts the wrong observable flips.

Knowing the per-qubit rates also helps at circuit level (Table 9 and Fig. ?? show no

Table 9: Circuit-level oracle gap on $[[72, 12, 6]]$ (6 rounds, $\eta = 20$, 5,000 log-normal $\sigma = 1$). n_{10} : shots per point (preliminary, $N = 5,000$ only the mean-prior decoder fails; n_{01} : shots, underpowered) only the oracle fails. The last column counts rate profiles on which the mean-prior decoder has more failures than the oracle / fewer.

p_{mean}	Shots	Mean-prior LER	Oracle LER	n_{10}/n_{01} (McNemar p)	Profiles
0.001	10,000	0 ($< 3.8 \times 10^{-4}$)	0 ($< 3.8 \times 10^{-4}$)	0/0	0/0
0.003	10,000	1.4×10^{-3}	0.6×10^{-3}	9/1 (0.027)	6/0
0.006	20,000	1.36×10^{-2}	0.60×10^{-2}	192/39 (1.5×10^{-23})	38/2

statistically detectable oracle gap at circuit level. With only 3 and 8 discordant pairs at the two tested operating points, however, McNemar’s test cannot distinguish a true zero gap from a gap of up to approximately 1.5% LER—consistent with the 8). At $p_{\text{mean}} = 0.006$ the oracle reduces the logical error rate from 1.36% to 0.60% (56%; McNemar $p = 1.5 \times 10^{-23}$), and it is better on 38 of 40 independent rate profiles (sign test $p = 1.5 \times 10^{-9}$), so the result is not driven by a few extreme profiles. At $p_{\text{mean}} = 0.003$ the gap has the same size (57%) but only 10 discordant shots, and at $p_{\text{mean}} = 0.001$ neither decoder fails in 10^4 shots. The circuit-level gap (56%) is somewhat smaller than the code-capacity oracle gap on the same code (65.1%), but it does not vanish.

Correction to an earlier version. An earlier version of this section reported logical error rates

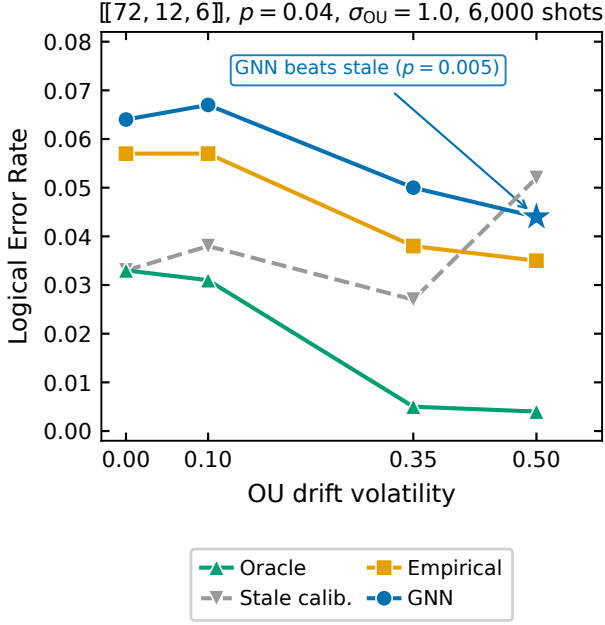


Figure 7: LER vs. OU drift volatility for four decoder configurations on $[[72, 12, 6]]$, $p = 0.04$, $\sigma_{OU} = 1.0$, 6,000 shots. Star marks the single operating point (vol=0.5) where GNN beats stale calibration ($p = 0.005$); GNN is beaten by the empirical frequency estimator at every volatility level.

of 0.27 and 0.34 at low contrast. The present sample cannot distinguish between negligible prior sensitivity and a nonzero gap below the resolution of the experiment. Higher shot counts (targeting ≥ 100 discordant pairs) are required to distinguish these cases. The two prior assignments are nevertheless numerically distinct: 86% of fault mechanisms change rank by more than 50 positions and the per-mechanism probability ratios span $0.25 \times 5.1 \times$. The DEM contains 3 $p_{\text{mean}} = 0.001$ and 0.003 and no oracle gap. Those circuits lacked the first-round and final detectors (360 instead of 432), so any fault that flipped a logical near the time boundaries was invisible to the decoder. This produced a logical error floor that no prior could remove. With the complete detector set, uniform noise at $p = 0.001$, 821 fault mechanisms and 360 detectors 0.003, suggesting that many fault mechanisms map to the same detector information. This high degeneracy is a plausible explanation for the weak prior sensitivity, but the present experiment does not isolate it causally. An ablation that changes DEM degeneracy while holding the prior mismatch fixed would be required for that conclusion.

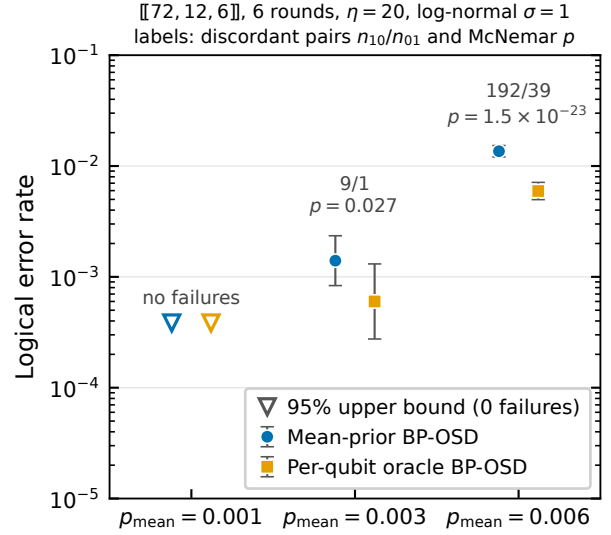


Figure 8: Circuit-level oracle gap on $[[72, 12, 6]]$, 6 rounds, $\eta = 20$, 5,000 shots. No statistically detectable difference between points: logical error rate with 95% Wilson intervals; open triangles: 95% upper bound when no failures were observed. The per-qubit oracle has 56% fewer failures than the mean-prior and per-qubit-oracle BP-OSD decoder at either operating point (McNemar $p \geq 0.25$; 3 and 8 discordant pairs—underpowered, see text) $p_{\text{mean}} = 0.006$.

Thus, the circuit-level experiment is currently a null result with insufficient statistical resolution; it does not establish that calibration side information is irrelevant at circuit level. 0.005 and 0.01 gives $0/10^4$, $5/10^4$, $14/10^4$ and 2.1×10^{-2} ; the legacy circuit gives 0.124 and 0.336 at $p = 0.001$ and 0.003.

6 Discussion

6.1 The GNN's role and its limits

TANNERGNN achieves a 15.4% LER reduction over the reduces the LER of the 10-iteration flooding BP decoder used in its training loop. The paired held-out test supports a real difference for this trained checkpoint, while gradient diagnostics show that the network produces non-trivial corrections. Robustness to training initialization has not yet been measured, so we do not treat the single-seed result as a reproducibility claim by $12.5\% \pm 0.6\%$ over five training seeds, and the gain persists outside the training range. The mechanism is mainly convergence: the GNN roughly halves the number of shots on which BP fails to reach a

syndrome-consistent estimate (1,740 to 838–975 of 20,000). Running BP for 100 iterations achieves more than this, and BP-OSD more still, so the result is best read as a cheaper route to part of the benefit of longer decoding, not as an improvement on state-of-the-art BB decoding.

The Regime-B result ~~is also informative: the learned correction does not transfer automatically to serial~~ follows from the failure-set structure (Table 6): BP-OSD. ~~This is consistent with the training/evaluation mismatch, but the residual failure sets have not yet been characterized well enough to claim a trapping-set mechanism’s failures are a near-subset of BP’s, and the GNN’s repairs fall mostly on shots BP-OSD already decodes. When passed to BP-OSD, the learned corrections reshuffle failures among the hardest shots and make OSD choose heavier incorrect estimates, giving a small net loss.~~ Training against ~~BP-OSD directly is complicated by the non-differentiable OSD stage; differentiable surrogates, distillation, or training the target decoder, for example through a differentiable surrogate of OSD or on the residual failures of the target decoder are possible alternatives~~BP-OSD, is the appropriate next test.

6.2 Benchmark design implications

The baseline experiments identify two reporting requirements for learned QLDPC decoders. First, BP-OSD schedule, update rule, iteration count, and OSD configuration must be stated explicitly because they can change LER by more than two orders of magnitude on the same code. Table 2 records these settings for the present study.

Second, the training and evaluation decoders must be distinguished. An LLR correction learned through flooding BP need not remain beneficial when the schedule, iteration count, and OSD stage are changed. Comparisons across such decoder configurations should identify the mismatch rather than treating all BP-based baselines as equivalent.

6.3 Comparison with prior work

Astra [5] uses a learned Tanner-graph message-passing decoder and reports BB-code performance competitive with, and in several settings better than, BP-OSD under code-capacity depolarizing noise. This places our ~~15.4%–12.5%~~

flooding-BP gain in context: improvement over plain BP is not, by itself, the main novelty. Our contribution is instead the controlled comparison showing how that gain changes when the baseline and side information are strengthened.

Gong et al. [4] interleave differentiable BP stages with GNN corrections on the same sparse graph and show error-floor improvements over several post-processing approaches. Our interleaved variant follows the same broad design principle, but ~~the checkpoint reported here does not train successfully. We therefore do not claim a competitive result~~ its decoding performance is not evaluated here, so we make no claim for that architecture.

Recent circuit-level work on BB codes also reports neural decoders that can outperform BP-OSD on the $[[72, 12, 6]]$ code while retaining a run-time advantage, although scaling to $[[144, 12, 12]]$ remains more challenging [14]. This comparison reinforces the need to evaluate learned decoders against a fully specified competitive non-learned baseline, not only against flooding BP.

Stein et al. [15] condition a neural QEC decoder on spatially resolved hardware calibration data using FiLM and evaluate on IBM repetition-code experiments. Their conditioning signal is per-qubit and calibration-derived, unlike the single scalar $\hat{p}$ used in our initial model. This distinction is consistent with the oracle experiments of Sec. 5.5, which motivate spatial rather than purely global conditioning.

6.4 Towards spatially-resolved conditioning

The oracle-gap experiment (Sec. 5.5) shows that per-qubit rate side information can materially improve the tested BP-OSD decoder. Because operating point and code size co-vary, the current data do not establish how this value scales with distance. Proposition 1 also shows that a uniform rescaling of the min-sum BP prior cannot reproduce the effect of spatially varying reliabilities. These results motivate testing spatially resolved FiLM (SRCF), in which per-qubit parameters (γ_i, β_i) are conditioned on persistent hardware calibration data rather than on a single graph-wide scalar. Such a decoder should be compared with the non-learned frequency estimator of Sec. 5.6 and evaluated ~~in a circuit-level setting where the calibration signal remains identifiable. Neither condition has at circuit level, where the oracle gap~~

remains large (56% at $p_{\text{mean}} = 0.006$, Sec. 5.7). Whether such rates can be estimated well enough from syndrome history has not been established in the present experiments.

6.5 Limitations

- (i) ~~The primary 15.4% GNN result uses one training seed. Statistical uncertainty over independent training runs has not been measured.~~
- (ii) ~~All GNN variants were trained against GNN was trained only through 10-iteration flooding BP. No variant was trained Training against serial BP-OSD, which is the competitive classical baseline. Whether GNN augmentation of serial BP-OSD yields gains, is untested.~~
- (iii) GNN checkpoints exist only for [72, 12, 6]; the larger codes were not trained ~~due to~~ because of CPU-only compute ~~constraints.~~ The null result in Regime B. The Regime-B result should not be extrapolated to the larger codes.
- (iv) ~~The interleaved model reached its best validation checkpoint at epoch 1, after which the validation loss increased. The current run is therefore insufficient to evaluate the interleaved architecture.~~
- (v) ~~All training and evaluation used CPU only, which limited hyperparameter search and training duration~~ them.
- (vi) The single per-qubit correction Δ_i ~~is~~ δ_j is trained on Z errors and applied to both $\Delta_{z,i}$ and $\Delta_{x,i}$ CSS components, which is sub-optimal under $\eta = 20$ ~~$\eta = 20$~~ biased noise; separate Z ~~and~~ X readout heads are the ~~recommended~~ natural fix.
- (vii) ~~Code-capacity noise omits measurement errors, leakage, and correlated faults present in real circuits. The interleaved architecture and the FiLM variant are described but their decoding performance is not evaluated.~~
- (viii) The circuit-level study uses one code, 6 rounds and a biased single-qubit channel after each gate; two-qubit correlated faults, leakage and crosstalk are not modelled, and

the paper's operating points $p_{\text{mean}} \leq 0.003$ are too clean to resolve the gap.

- (ix) Hyperparameter search and training duration were limited by CPU-only compute.

7 Conclusion

On [72, 12, 6], a Tanner-graph GNN trained through 10-iteration flooding BP reduces ~~the held-out LER by 15.4% that decoder's LER by $12.5\% \pm 0.6\%$ at $p = 0.04$ and $\eta = 20$. The same learned correction does not improve the serial min-sum $10.6\% \pm 0.7\%$ at $p = 0.06$, reproducibly over five training seeds and on leakage-free held-out data. The gain comes mostly from helping BP converge. It does not reach 100-iteration BP or serial BP-OSD~~ reference decoder and increases LER at the two highest tested error rates. The demonstrated gain is therefore specific to the training-aligned flooding-BP baseline; the present experiments do, and the same corrections make BP-OSD slightly worse because the shots they repair are largely ones BP-OSD already decodes. An improvement over one BP baseline therefore does not establish an improvement over competitive BB decoding more generally a stronger decoder.

~~The baseline experiments show that BP schedule, update rule, iteration budget, and OSD configuration can change finite-length LER by orders of magnitude. These parameters must be treated as part of the decoder specification when learned and non-learned methods are compared. We also Decoder configuration matters as much as the learned component: on [288, 12, 18], schedule and OSD post-processing alone change LER by about 47%. We proved that uniform positive rescaling of the channel LLRs leaves the hard decisions of ideal unclipped min-sum BP unchanged. A global prior magnitude alone therefore cannot provide spatial adaptation to heterogeneous qubit reliabilities, although a nonlinear learned model may still use, so a global noise estimate as context.~~

~~The synthetic oracle experiments identify per-qubit calibration as the more informative adaptive signal in the tested setting; revealing the cannot by itself make min-sum BP adaptive. Per-qubit calibration, however, is highly informative: revealing true per-qubit rates~~

~~substantially improves reduces BP-OSD under heterogeneous's LER by up to 88% at code capacity and by 56% under circuit-level noise. The current cross-code experiments do not isolate how this gain scales with distance, and the circuit-level study does not yet have sufficient statistical resolution. A direct next test is to condition a learned decoder conditioned on persistent spatial calibration data and compare it, compared against a fully specified serial BP-OSD baseline and the a non-learned frequency estimator, using multiple training seeds and a noise model that preserves temporal calibration structure. per-qubit estimator.~~

Acknowledgments

The authors thank the STAR Lab program at the University of Arizona for facilitating this research and providing computational resources.

Data and code availability

Source code, trained checkpoints, ~~all datasets, and the audit scripts that produced the findings of Secs. 5.1 and 5.2 are available at . Syndrome data were generated using datasets and analysis scripts are available at https://github.com/kondshk/adaptive_gnn. All tables and figures except the drift experiment (Sec. 5.6) are produced by a committed script from stored per-shot outcomes: `audit_headline_v2.py` (headline result), `audit_tables_v2.py` (baseline, invariance, BP-OSD and oracle tables), `audit_failure_sets.py` (failure sets) and `audit_circuit_v2.py` (circuit level); `docs/PROVENANCE.md` maps each reported number to its source. Code-capacity data are sampled directly with NumPy; circuit-level data are generated with Stim [16].~~

References

- [1] Sergey Bravyi, Andrew W. Cross, Jay M. Gambetta, Dmitri Maslov, Patrick Rall, and Theodore J. Yoder. “High-threshold and low-overhead fault-tolerant quantum memory”. *Nature* **627**, 778–782 (2024).
- [2] IBM Quantum. “IBM lays out clear path to fault-tolerant quantum computing”. IBM Quantum Computing Blog (2025).
- [3] Joschka Roffe, David R. White, Simon Burton, and Earl Campbell. “Decoding across the quantum LDPC code landscape”. *Physical Review Research* **2**, 043423 (2020).
- [4] Anqi Gong, Sebastian Cammerer, and Joseph M. Renes. “Graph neural networks for enhanced decoding of quantum LDPC codes”. In Proc. IEEE International Symposium on Information Theory (ISIT). (2024).
- [5] Arshpreet Singh Maan and Alexandru Paler. “Machine learning message-passing for the scalable decoding of QLDPC codes”. *npj Quantum Information* **11**, 78 (2025).
- [6] A. Robert Calderbank and Peter W. Shor. “Good quantum error-correcting codes exist”. *Physical Review A* **54**, 1098 (1996).
- [7] Andrew M. Steane. “Error correcting codes in quantum theory”. *Physical Review Letters* **77**, 793 (1996).
- [8] Marc P. C. Fossorier and Shu Lin. “Soft-decision decoding of linear block codes based on ordered statistics”. *IEEE Transactions on Information Theory* **41**, 1379–1396 (1995).
- [9] P. V. Klimov et al. “Fluctuations of energy-relaxation times in superconducting qubits”. *Physical Review Letters* **121**, 090502 (2018).
- [10] Malcolm Carroll, Sami Rosenblatt, Petar Jurcevic, Isaac Lauer, and Abhinav Kandala. “Dynamics of superconducting qubit relaxation times”. *npj Quantum Information* **8**, 132 (2022). [arXiv:2105.15201](https://arxiv.org/abs/2105.15201).
- [11] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. “FiLM: Visual reasoning with a general conditioning layer”. In Proc. AAAI. Volume 32. (2018). [arXiv:1709.07871](https://arxiv.org/abs/1709.07871).
- [12] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. “Focal loss for dense object detection”. In Proc. IEEE ICCV. Pages 2999–3007. (2017).
- [13] Quinn McNemar. “Note on the sampling error of the difference between correlated proportions or percentages”. *Psychometrika* **12**, 153–157 (1947).
- [14] John Blue, Harshil Avlani, Zhiyang He, Liu Ziyin, and Isaac L. Chuang. “Machine learning decoding of circuit-level noise for bivari-

- ate bicycle codes”. [Quantum](#) **10**, 2149 (2026).
[arXiv:2504.13043](#).
- [15] Samuel Stein, Shuwen Kan, Chenxu Liu, Adrian Harkness, Sean Garner, Zefan Du, Yufei Ding, Ying Mao, and Ang Li. “Calibration-conditioned featurewise linear modulation decoders for low-latency decoding of quantum error correction evaluated on IBM repetition-code experiments”. [Physical Review Research](#) **8**, 033194 (2026).
[arXiv:2601.16123](#).
- [16] Craig Gidney. “Stim: A fast stabilizer circuit simulator”. [Quantum](#) **5**, 497 (2021).